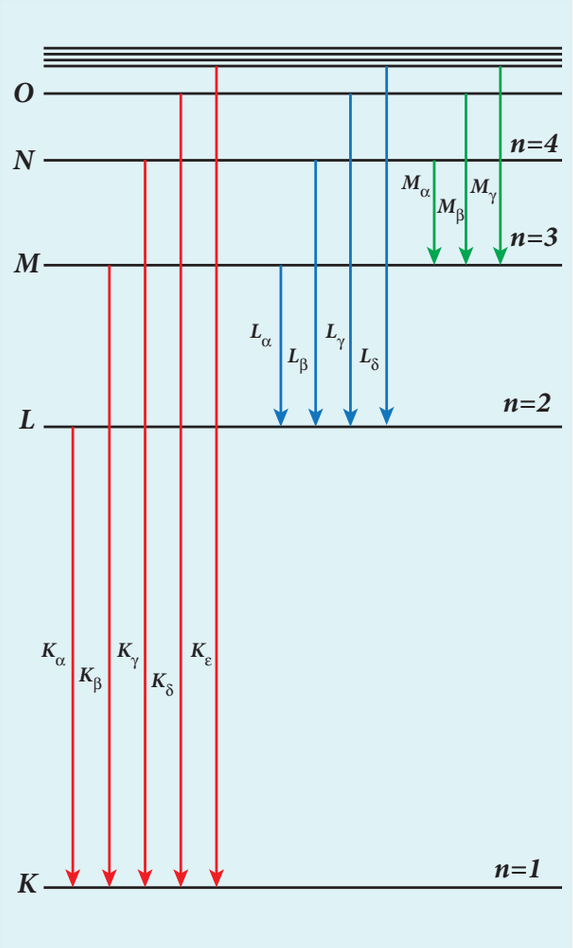


Figure 8.23 Origin of characteristic x-ray spectra

electronic transitions from *M*, *N*, *O* level to the L-level and so on.

The K_α and K_β of the *K*-series of molybdenum are shown by the two peaks in its x-ray spectrum in Figure 8.21(b).

Applications of x-rays:

X-rays are being used in many fields. Let us list a few of them.

1) Medical diagnosis

X-rays can pass through flesh more easily than through bones. Thus an x-ray radiograph containing a deep shadow of the bones and a light shadow of the flesh may be obtained. X-ray radiographs are used to detect fractures, foreign bodies, diseased organs etc.

2) Medical therapy

Since x-rays can kill diseased tissues, they are employed to cure skin diseases, malignant tumours etc.

3) Industry

X-rays are used to check for flaws in welded joints, motor tyres, tennis balls and wood. At the custom post, they are used for detection of contraband goods.

4) Scientific research

X-ray diffraction is important tool to study the structure of the crystalline materials – that is, the arrangement of atoms and molecules in crystals.

EXAMPLE 8.9

Calculate the cut-off wavelength and cut-off frequency of x-rays from an x –ray tube of accelerating potential 20,000 V.

Solution

The cut-off wavelength of the x-rays in the continuous spectrum is given by

$$\lambda_\circ = \frac{12400}{V}\text{Å} = \frac{12400}{20000}\text{Å}$$
$$= 0.62\text{Å}$$

The corresponding frequency is

$$\nu_0 = \frac{c}{\lambda_0} = \frac{3\times10^8}{0.62\times10^{-10}} = 4.84\times10^{18}\,\text{Hz}$$

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